

BROOKHAVEN CALABRO, NY, USA

WMO: 725016

Lat: 40.822N

Lon: 72.869W

Elev: 82

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54790

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	14.8	-5.5	4.1	14.7	-1.2	5.2	18.2	26.3	31.7	24.2	29.1	7.9	330	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	88.4	73.5	85.4	72.2	82.8	71.4	77.1	83.0	75.8	81.3	74.5	79.7	9.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	133.3	79.9	73.9	126.9	78.7	72.8	121.9	77.8	40.5	82.5	39.2	81.5	38.0	79.6	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.9	19.5	17.5	DB	3.3	94.2	5.1	3.3	-0.4	96.6	-3.4	98.6	-6.3	100.4	-10.0	102.8
(5)				WB	2.0	79.2	4.9	1.9	-1.5	80.5	-4.4	81.6	-7.1	82.7	-10.7	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	31.3	32.5	39.2	49.2	58.6	67.8	74.5	73.1	66.5	55.7	45.7	37.5	(6)
(7)		DBStd	16.66	9.65	8.51	8.23	7.11	6.66	6.20	4.75	4.99	6.02	7.83	7.90	8.53	(7)
(8)		HDD50	2126	580	491	346	97	7	0	0	0	33	175	397		(8)
(9)		HDD65	5282	1043	910	801	476	218	41	1	3	55	303	579	852	(9)
(10)		CDD50	3129	1	1	10	73	274	535	758	717	494	210	46	10	(10)
(11)		CDD65	811	0	0	0	2	21	125	294	254	99	16	0	0	(11)
(12)		CDH74	5399	0	0	0	29	120	755	2182	1822	450	41	0	0	(12)
(13)		CDH80	1318	0	0	0	7	19	190	613	427	57	5	0	0	(13)
(14)	Wind	WSAvg	7.5	8.6	8.7	9.1	8.5	7.2	6.9	6.5	5.9	6.0	7.1	7.6	8.1	(14)
(15)	Precipitation	PrecAvg	44.8	3.5	3.1	4.4	3.9	3.4	3.6	3.3	3.8	4.1	4.1	3.3	4.6	(15)
(16)		PrecMax	58.5	7.5	5.3	11.8	7.0	7.0	8.5	5.7	9.0	7.6	17.8	5.4	8.2	(16)
(17)		PrecMin	33.6	1.8	1.1	0.7	1.7	0.6	0.2	1.1	0.8	1.2	0.4	0.8	1.6	(17)
(18)		PrecStd	6.0	1.3	1.3	2.4	1.6	1.5	2.2	1.3	2.2	1.7	3.3	1.3	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	57.5	66.3	78.7	82.6	90.1	93.6	90.6	85.2	79.4	66.9	61.5	(19)
MCWB			53.6	51.8	54.8	59.9	68.1	74.0	75.5	74.6	74.1	68.2	60.2	56.6	(20)	
2%		DB	52.9	52.6	58.8	69.7	77.2	85.0	89.6	87.3	81.3	73.6	63.8	57.5	(21)	
		MCWB	50.6	48.1	48.4	55.8	64.4	70.7	73.7	73.5	69.6	67.9	58.8	54.9	(22)	
5%		DB	49.5	49.2	54.6	64.5	73.3	81.5	86.0	84.4	78.8	70.7	61.6	54.7	(23)	
		MCWB	46.1	44.9	48.1	53.1	62.6	69.3	72.8	72.3	68.8	64.3	56.9	52.0	(24)	
10%		DB	46.0	45.7	51.3	60.7	69.9	78.2	83.0	82.0	76.4	68.1	58.9	51.9	(25)	
		MCWB	41.9	41.3	45.9	51.1	60.9	67.7	71.9	71.8	68.3	62.9	54.7	48.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	53.7	57.5	64.0	71.2	75.9	78.7	79.2	75.9	71.9	63.5	58.9	(27)
MCD B			55.7	55.2	64.3	73.7	80.1	85.1	87.6	85.5	81.7	75.6	64.8	60.4	(28)	
2%		WB	50.9	49.6	52.4	59.1	67.9	73.7	77.1	77.5	73.9	68.8	60.6	55.5	(29)	
		MCD B	52.6	51.8	55.4	65.0	73.9	82.1	83.9	82.0	78.0	72.6	63.0	57.1	(30)	
5%		WB	46.5	45.9	49.5	56.2	65.5	71.8	75.9	76.1	72.4	66.5	58.1	52.6	(31)	
		MCD B	48.8	48.4	53.4	61.3	70.8	77.8	81.7	80.7	75.9	69.7	60.8	54.6	(32)	
10%		WB	42.8	41.9	46.3	53.5	63.1	70.1	74.7	74.6	70.8	63.9	55.5	48.7	(33)	
		MCD B	45.3	45.2	50.1	57.7	68.0	75.3	80.0	79.3	74.4	67.0	58.4	51.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.2	15.5	15.4	17.0	16.4	15.7	15.2	15.7	16.4	16.6	16.1	14.3	(35)	
5% DB			MCD BR	17.6	18.3	19.5	23.0	20.9	19.2	18.5	17.4	17.3	16.8	16.2	15.3	(36)
			MCWBR	16.5	15.3	13.6	12.2	10.7	9.0	7.3	7.6	9.6	11.1	13.2	15.5	(37)
		5% WB	MCD BR	16.9	17.9	17.2	17.5	17.2	15.9	14.1	12.6	12.6	13.5	14.2	14.9	(38)
MCWBR			16.8	16.8	14.2	11.8	10.0	8.5	6.8	6.6	8.5	10.9	13.4	16.0	(39)	
			Clear-Sky Solar Irradiance	taub	0.309	0.317	0.337	0.383	0.434	0.476	0.490	0.466	0.396	0.348	0.318	0.309
taud	2.517	2.494		2.479	2.369	2.262	2.186	2.155	2.217	2.383	2.531	2.562	2.560	(41)		
Ebn at Noon	271	286		291	282	269	257	252	254	267	271	266	263	(42)		
Edh at Noon	23	28		31	37	43	46	47	43	34	26	22	21	(43)		
(44)	All-Sky Solar Radiation	RadAvg	582	864	1196	1543	1725	1885	1908	1712	1395	955	660	496	(44)	
(45)		RadStd	49	83	100	158	169	182	99	107	110	88	54	43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	+110

Nomenclature: See separate page